Allen Saji

Coorparoo, QLD | 0421 437 800 | allensaji.com | allensaji012@gmail.com

SUMMARY

Final-year engineering student at QUT with a hybrid software and electrical background. Project experience spans AWS cloud architecture, React frontend development, embedded C systems, power electronics, and computer vision research. Open to internships and graduate roles in software or electrical engineering.

EDUCATION

Queensland University of Technology — Expected Early 2027
Bachelor of Engineering (Honours), Computer and Software Systems — Brisbane, QLD

- Minoring in Advanced Electrical Engineering and Robotics & Embedded Systems
- Dean's List

Indian School Al Ghubra — Jun. 2022
High School Diploma – 7.0 GPA — Muscat, Oman

- Executive, Robotics Club (2018 – 2020)

PROJECTS

Honours Research: Bubble Analysis in Water Electrolysers | *Python, PyTorch, Detectron2* — Feb. 2026 – Oct. 2026

- Built a computer vision pipeline that automatically outlines bubbles in 2,000 fps electrolyser video, training Mask R-CNN on a dataset of 37,388 labels audited from three annotators' work
- Improved segmentation accuracy (mask AP) from 40.4 to 45.1 by fine-tuning at a higher input resolution, reaching an F1 of 0.80, and compared the model with StarDist and a hybrid on held-out test data
- Detected 94–100% of bubbles above 25 px, including overlapping ones, and measured gas coverage and bubble size (Sauter mean diameter) to within about 3%; total counts ran about 20% low because very small bubbles were missed
- Measured how often human labellers agree and found the model's small-bubble limit (under about 15 px) lines up with where they stop agreeing, making a consistent labelling rule the top recommendation

Cloud-Based Background Removal System | *AWS, Node.js, Docker* — Aug. 2025 – Nov. 2025

- Designed and deployed a cloud-native background removal platform on AWS as two independent containerised services: a stateless API tier and an ML processing tier, orchestrated with ECS Fargate for serverless compute and auto-scaling
- Architected an asynchronous SQS pipeline to decouple user requests from ML inference, with retry logic and a Dead Letter Queue to automatically isolate failing jobs after three attempts
- Integrated a neural network-based background removal model into the processor service, with processed images stored in S3 and job status tracked through PostgreSQL on RDS
- Secured the platform with AWS Cognito authentication and S3 pre-signed URLs so image data bypassed the API servers entirely, keeping the compute tier stateless and horizontally scalable; deployed behind an ALB with HTTPS and a custom domain

Portfolio Website | *React 19, Vite, Framer Motion, CSS* — Dec. 2025 – Present

- Designed and built allensaji.com end-to-end, developing a custom CSS design-token system that powers synchronised light and dark themes with no flash on page load
- Engineered an interactive project timeline component with distinct desktop and mobile interaction models: click-to-expand detail panels with hover previews on desktop, and independent tap-to-toggle cards on mobile that adapt seamlessly as the viewport changes
- Built a shared animation system on top of Framer Motion to drive scroll reveals, staggered transitions, and panel animations consistently across the site, with full support for users who prefer reduced motion
- Made the site usable across devices and preferences with responsive layouts, light and dark themes, and full keyboard navigation; deployed on Vercel

Solar Inverter Reverse Engineering | *LTspice, MATLAB* Feb. 2025 – Jun. 2025

- Reverse-engineered a commercial 5 kW grid-tied PV inverter with a partner, reconstructing both the DC-DC boost stage and single-phase full-bridge DC-AC stage from physical measurements and datasheet specifications
- Modelled the boost converter in LTspice and MATLAB, validating theoretical ripple and duty cycle calculations against simulation and estimating stage efficiency after accounting for inductor, switching, and capacitor losses
- Designed the full-bridge inverter stage using sinusoidal pulse-width modulation to push switching noise above the audible range, selecting a suitably rated IGBT and sizing the output filter to meet grid voltage and waveform quality requirements
- Proposed a snubber circuit to significantly reduce switching losses in the boost MOSFET, and identified synchronous rectification as a further efficiency improvement path

Elevator System Simulation | *C, POSIX, TCP/IP* Aug. 2024 – Oct. 2024

- Designed and implemented a multi-component elevator system in C for Linux using POSIX shared memory, multithreading, and TCP/IP sockets
- Built modules for car control, scheduling, safety, and user interaction with emphasis on inter-process communication and real-time responsiveness
- Followed safety-critical standards (MISRA C) and developed a floor scheduling algorithm integrated via a custom build system

EXPERIENCE

Coles May 2024 – Present
Team Member (Service and Online) Brisbane, QLD

- Led day-to-day operations within the online team, coordinating workflows to ensure efficient order fulfilment and high customer satisfaction
- Acted as a first point of contact for customer issues, resolving disputes and escalating when necessary to maintain service quality
- Trained and mentored new team members, improving team performance, consistency, and adherence to procedures
- Managed high-volume transactions using point-of-sale systems with accuracy and efficiency, maintaining smooth store operations

Wildfire Sports & Trek Oct. 2023 – Feb. 2024
Warehouse Dispatcher Brisbane, QLD

- Reduced order processing times with streamlined picking, packing, and shipping procedures
- Participated in continuous improvement initiatives to streamline workflows and optimise productivity
- Prepared shipments for safe transportation by securely wrapping parcels and labelling packages according to company and shipping guidelines

SKILLS

Programming : Python, C, C#, JavaScript/Node.js, HTML/CSS
Frontend : React, Vite, Framer Motion
Cloud : AWS (ECS Fargate, EC2, SQS, S3, RDS, Cognito, ALB, CloudFront), Docker, Git, Linux (WSL)
Databases : PostgreSQL, REST APIs
Engineering & Simulation : MATLAB/Simulink, LTspice, KiCad/Eagle, AutoCAD
Electrical & Power Systems : DC-DC converters, PV inverter modelling, PCB design & layout, circuit debugging
Embedded Systems : Microcontroller programming (C), real-time systems, UART/SPI/I2C, hardware debugging
Machine Learning & Computer Vision : PyTorch, Detectron2, OpenCV, TensorFlow/Keras, image processing, instance segmentation, neural network integration
Languages : English (Native), Malayalam (Native)